

Plug and Play Conversations: The Micro-Conversation Scheme for Modular Development of Hybrid Conversational Agent

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ABSTRACT

Conversational agents (CAs) for psychotherapy pose unique challenges (e.g., reliance on domain experts to pre-script large amounts of therapeutical dialogues). To tackle these challenges, we propose a modular approach to develop such CA, called Micro-Conversation Scheme (MCS). Conversations can be algorithmically extended in MCS by combining different micro-conversations (MC), which isolate single therapeutical topic. The sequencing of MC is managed by a Connector component, connecting MC into longer conversations with context. Additionally, MCS integrates natural language generation (NLG) models as plugins for generating counseling-style utterances (e.g., reflections). Moreover, MCS adopts interactive learning to continuously improve CA based on human feedback. MCS provides a solution to the challenges of complex-to-design and difficult-to-extend conversations, and inability of CA to flexibly generate context-appropriate counseling-style utterances for psychotherapy. MCS is expected to benefit the community by promoting the collaboration between conversational designers and developers while preserve their independence during the development of CAs.

CCS CONCEPTS

• **Human-centered computing** → **User interface toolkits; User interface toolkits; Computer supported cooperative work.**

KEYWORDS

Collaborative development, modular and interactive development, hybrid conversational agent

ACM Reference Format:

Xin Sun, Emiel Krahmer, Jan de Wit, Reinout Wiers, and Jos A. Bosch. 2023. Plug and Play Conversations: The Micro-Conversation Scheme for Modular Development of Hybrid Conversational Agent. In *Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*, October



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CSCW '23 Companion, October 14–18, 2023, Minneapolis, MN, USA

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ACM ISBN 979-8-4007-0129-0/23/10.

<https://doi.org/10.1145/3584931.3606998>

14–18, 2023, Minneapolis, MN, USA. ACM, New York, NY, USA, 6 pages.
<https://doi.org/10.1145/3584931.3606998>

1 INTRODUCTION

Conversational agents (aka chatbots) are increasingly used in psychotherapy and behavioral intervention counseling [1, 10, 32]. Most of these chatbots utilize pipeline-based dialogue systems [4], concentrating on specific tasks during psychotherapy or counseling. Contrary to these task-oriented chatbots led by users, counseling chatbots need to guide conversations, resembling human counselors' roles in therapeutic interactions. Accordingly, most counseling chatbots necessitate domain experts acting as conversational designers to pre-script extensive dialogues, including counseling-style utterances like reflections. [1, 22]. To facilitate collaboration between domain experts (as designers) and developers in creating CAs grounded in theories of psychology and behavioral intervention, we devised a modular and interactive approach called the Micro-Conversation Scheme (MCS). MCS simplifies chatbot development by modularly designing multiple brief and topic-specific pipelines ('micro-conversations'), minimizing the reliance on domain experts to pre-script dialogues for the entire counseling session. The Connector facilitates personalization and expansion of micro-conversations, while the NLG component allows for flexible generation of diverse counseling-style and transition utterances. Moreover, an interactive component in MCS enables individuals to give feedback on conversations that can continuously improve components in MCS. Therefore, the primary **contributions and novelties** of this work include: **1). Modularization:** MCS facilitates collaboration between CA designers and developers while preserving independency in CA development, making the process more agile and extensible; **2). Hybrid:** The NLG component in MCS introduces generative capability to pre-scripted conversations enabling the flexible generation of counseling-style utterances; **3). Extensibility:** MCS enables conversations to be expanded and automatically ordered by algorithms, providing less rigidity compared to fully pre-scripted conversations; **4). Improvability:** MCS allows the continuous improvement of involved components based on the human feedback collected during the interaction.

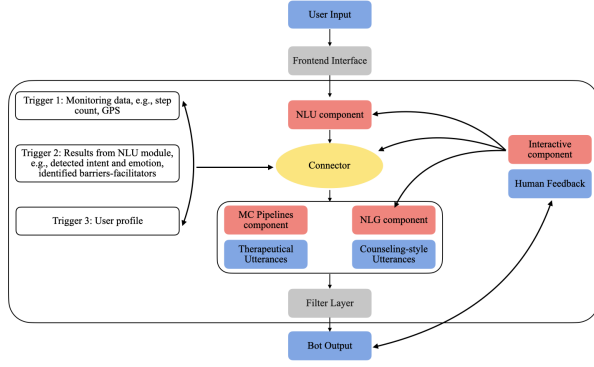


Figure 1: The MCS-based CA architecture consists of a frontend interface for user input, a multi-task NLU component for intent and trigger classification, a Connector to manage corresponding MCs, and a filtering layer to determine bot output from therapeutic MC pipelines component or NLG component. Additionally, human feedback on bot output enables continuous improvement of other components.

2 RELATED WORK

2.1 Conversational agents in psychotherapy and behavior intervention

CAs (or chatbots) for psychological counseling and lifestyle interventions provide cost-effective health solutions. Research [10, 15, 34] demonstrates their effectiveness in promoting positive behavior through chatbot delivered interventions, especially when combined with wearable devices [9, 24]. Current methods heavily rely on domain experts and pre-scripted conversations limiting diversity and extensibility. However, counseling chatbots should not only have pre-scripted therapeutic conversations and dialogue flow but also have counseling-style utterances (i.e., the empathetic, reflective, or non-directive utterances used by therapists to facilitate client self-exploration) [28, 33]. We aim to develop a chatbot capable of providing therapeutic utterances based on domain knowledge and also flexibly generating counseling-style utterances.

2.2 Design hybrid conversational agent with NLG and interactive learning

The present work is mostly based in Motivational Interviewing (MI) [16], a technique applicable to various psychotherapies. Previous research has explored the development of theory-grounded chatbots for behavioral intervention or psychotherapy [1, 5, 7, 8, 19, 29, 30], affirming the effectiveness of such chatbots. However, the design and development of such chatbots demands significant investment from domain experts and developers to craft comprehensive dialogues for the entire therapy session, which results in a labor-intensive challenge in this field and restricts the flexibility and extensibility of such chatbots. Prior studies [12, 27] attempted to automate MI conversations focusing on counseling-style utterances generation (i.e., reflection and empathy). However, counseling-style responses from most MI-based CA are manually pre-scripted by

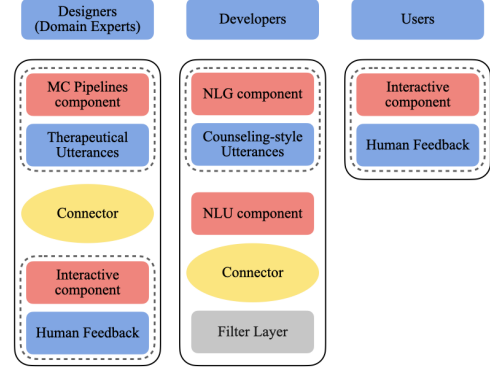


Figure 2: Designers, developers and users can collaboratively but independently contribute to various MCS components. Designers pre-script therapeutic MC pipelines and topic relationships as domain knowledge, which developers use to implement models in Connector, NLU, NLG components. Everyone can provide input to the interactive component, and all components are subsequently assembled into a conversational agent.

domain experts. Recent research [27, 28] explores automatic generation of counseling-style utterances using large language models (LLMs) and few-shot method. However, assessing quality of LLMs generation remains challenging due to subjectivity and context-dependence, rendering metrics like BLEU [21] and ROUGE [11] scores insufficient. Reinforcement Learning from Human Feedback (RLHF) [13, 17, 18] offers a solution that using human assessment and reinforcement learning to continuously optimize language models and better align them with human values.

3 MICRO-CONVERSATION SCHEME

Overall, MCS comprises five distinct but connected components to facilitate close collaboration between conversational designers and chatbot developers (refer to Figures 1 and 2): a multi-task NLU component, multiple micro-conversational pipelines (MC pipelines), a Connector component, a NLG plugin component and an interactive component.

3.1 Multi-task NLU component

Multiple classification models can be integrated into the multi-task NLU component, each model performing different classifications that allow CA to produce personalized and context-aware conversations. For instance, we firstly implement an intent classification model in the multi-task NLU component, which allows CA to understand user's intent and return proper responses. The second classification model is an emotion detection model. We use the pre-trained model from Huggingface [31] and perform dynamic emotion detection on conversations. CA can trigger mental diagnostics if user frequently reflects negative emotions. The multi-task NLU component can be extended by adding more classification models in the future as requested by conversational designers. Additionally,

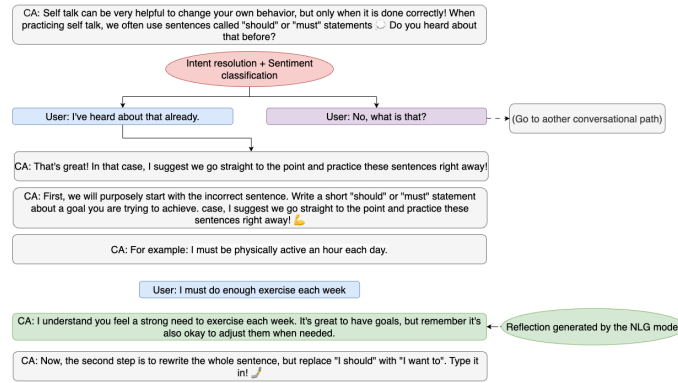


Figure 3: Conversational snippet to promote PA. Grey indicates CA’s output, user inputs are in purple or blue. The red oval highlights multi-tasks NLU component like intent and sentiment classification, while the green signifies the NLG component for reflections.

the multi-task NLU component can process triggers from monitoring devices as specific intents, improving the context-awareness and personalization of CA.

3.2 Micro-conversational pipelines component

The application of this work presents a counseling CA that leverages MI and SDT theories to encourage healthy behavior adoption. Each MC pipeline corresponds to a topic for discussion during counseling (Example MC in Figure 3), allowing for brief conversations, similar to how human dialogues naturally shift between various subjects. For example, separate MC pipelines are used to address users’ confidence in increasing physical activity (PA) or make plans for behavioral change. Individual domain experts (designers) can independently design each MC pipeline without considering how their own designed MC integrate with those from other designers, leading to more extendable conversations. Consequently, MCS reduces complexity and time consumption in conversational design while enhancing extensibility compared to previous approaches [1, 6], as MC pipelines can be expanded independently without redundant work.

3.3 Connector component

The role of Connector in MCS is to automatically combine various MC pipelines, removing the need for domain experts to pre-script extensive dialogues for the entire counseling session, thereby reducing complexity and time investment in conversational design. The Connector comprises two elements: a connecting model and a transition action. The connecting model is used to predict the next MC pipeline during counseling based on dialogue history and user context, which is conceptually similar but completely different from prior work such as finite state machines [3, 26]. The transition action represents the transitive utterances between two consecutive MC pipelines. The Connector can predict the next MC and transition to that MC pipeline through various triggers, including: 1) receiving data from monitoring devices, 2) multi-task NLU module results suggesting a transition, 3) user profile and context initiating another MC pipeline, or 4) completion of the previous pipeline. This set of triggers can be extended by adding more classification

models to the multi-task NLU component or granting the CA access to additional monitoring data.

In this work, we designed 12 therapeutic MC pipelines (Example MC in Figure 3) to promote PA. Designers advised on all connectable MC pipeline pairs among these 12 MC pipelines. Subsequently, various sequences of MC pipelines were automatically generated by an algorithm called ‘Deep Walk’ [23]. In each connectable MC pipeline pair given by designers, MC pipelines are considered as nodes, with the prior node representing the previous MC pipeline and the post node denoting the next MC pipeline. Sequences generated from “Deep Walk” provide the priming data for building an initial connecting model to predict the next MC pipelines during testing phase. Actual MC pipeline sequences are collected through the conversational interaction, with the initial connector model from ‘DeepWalk’ offering three options for users to select (see Figure 4). The interactions create more actual sequences which can be used to train the ultimate connecting model (i.e., an RNN model for sequential prediction) within Connector for producing coherence MC pipeline sequences. As results, connecting model can form larger conversations by linking MC pipelines without relying on designers to pre-script large amounts of conversations spanning the entire counselling session.

3.4 NLG component

MCS incorporates a NLG component as plugin for producing diverse counseling-style conversations without pre-scripting. NLG component can automatically and flexibly generates essential counseling -style utterances, such as reflections in MI (as shown in Figure 5), which reduces design complexity while maintaining conversational flexibility. Besides, NLG component can be used to generate the transition utterance between two consecutive MC pipelines (which is the element in Connector introduced in section 3.3) as well. For instance, if a MC pipeline discusses daily step count and the next MC pipeline is about goal setting, the generated transition utterance could be: “user name, we talked about your daily step count, which helped me understand your PA status better. Shall we discuss setting a realistic daily step goal for next week?” This smooth transition facilitates conversational extensibility in

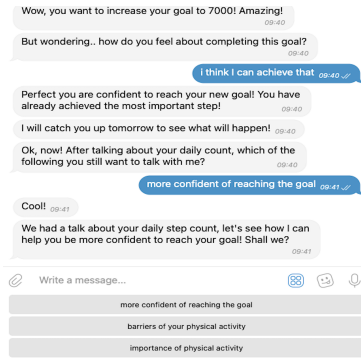


Figure 4: Interaction to collect actual sequences of MC pipelines.

MCS, reducing the need for designers to manually pre-script transitions between all different combinations of two consecutive MC pipelines.

In NLG component, we utilize few-shot generation with LLMs [14, 25], such as GPT-2 and GPT-3. These models are easily accessible through Huggingface library [31] or OpenAI API. The input prompts for NLG component are based on prior work [27]. However, in order to avoid the generated utterances from NLG component impacting conversational flow (e.g., generated question might lead pre-scripted conversations into unexpected and uncontrollable route) [33] or generating harmful content, we filter out generated questions and inappropriate utterances using an extra classification model as the filtering layer which ensures that only statement responses and safe utterances can be output, making it is possible for CAs to adopt generative models in sensitive domains (e.g., psychotherapy).

3.5 Interactive component

MCS enables continuous improvement of other components through an interactive component. For Connector, the initial connecting model can generate three options of next MC pipelines when triggered as mentioned in section 3.3, allowing users to select their preferred topic for further talk (see Figure 4). This interaction helps collect more actual MC pipeline sequences which serve as training data for the continuous improvement of ultimate connecting model (i.e., an RNN model). Moreover, we provide users autonomy to give feedback on utterances generated by NLG component. The feedback can serve as training data to continually enhance performance of NLG component using the method RLHF [13, 18]. Human feedback can take various formats, such as normal users to rate generated utterances using "thumb up, thumb down, or neutral" actions, while domain experts can offer more domain-specific feedback on generated utterances or ongoing dialogues based on their expertise. Through this interaction, we can not only collect human feedback to expand the conversational corpus but also improve NLG component by incorporating human feedback.

Prompt:

CA: What do you think about increasing your goal with 1000 steps, which is about an extra 7 minutes of walking? Do you think that would be feasible?

User: No, not really.

CA: Alright, tell me which number do you think is an achievable step goal for you?

User: Maybe same steps as this week

Generation of reflection from the NLG model:

CA (Reflection): I understand that increasing your step count right now may feel challenging. It's important to set goals that feel attainable for you. Keeping your steps the same as this week sounds like a comfortable pace for you at this moment. We can revisit this in the future when you feel ready!

Figure 5: Prompts and the generated counseling-style utterance.

4 PRELIMINARY USER STUDY

4.1 Pilot study

A pilot study was conducted with a sample of general population in August 2022 with two objectives: 1). to collect more conversational corpus data; 2). to get inspiration and feedback for therapeutical conversation design.

Participants: 30 participants were recruited with average age 24.63(SD=8.39,Min=18,Max=65), of which 7 were males and 23 were females. 6 participants has a high school degree, 14 has a bachelor degree and 10 has a master degree. Participants' average chatbot experience level on scale from 1 (no experience) to 5 (lots of experience) was 2.47(SD=0.97).

Procedure: The lab-based study had participants interact with a chatbot on their smartphone and completed survey afterward. The participants were firstly required to sign informed consent and information letter, and then were instructed to install Telegram (the frontend interface of our chatbot) on their smartphone. After that, participants were led to a cubicle with a computer and told to read the instructions on Qualtrics. They then completed demographic questionnaire and were provided with context for subsequent conversations with the chatbot and questionnaires.

Findings: Results revealed that chatbot was rated as effective with means of all three questionnaires higher than average(M=3.00). The effectiveness of chatbot for behavioural change is rated as 3.15 out of 5 by self-constructed questionnaire. Results from BUS-15 (Bot Usability Scale) [2] indicated that chatbot was quick to respond, accessible and easy to understand. The adapted WAI-SR (Working Alliance Inventory-Short Form Revised) [20] measures agreement on chatbot tasks, therapy goals and emotional bonds, and the scores of sub-scales were all above average. However, improvements are needed in conversation depth, human likeness, interactive methods (e.g., button or free text input) and personalized conversations. Clearer explanations when it doesn't understand user input, increased flexibility in unfamiliar situations and expanding the corpus should be helpful in addressing these issues.

Questionnaires	Mean	SD
Effectiveness Rating	3.15	0.87
BUS-15	3.35	1.36
Adapted WAIS-SR	3.27	1.02

Table 1: Mean and SD of used questionnaires.

4.2 Interview on conversational designer and developer

The qualitative interview involved six conversational designers and a developer aiming to assess the effectiveness of MCS in fostering collaboration and streamlining the CA development. Overall, interview demonstrated MCS's effectiveness in fostering collaboration and streamlining development. It also provides flexibility, extensibility, continuous improvement and generative capability to CA, which are favorable by designers and developer. Key findings are as follows:

1). Facilitated Collaboration: Most designers experienced improved coordination with developer and other designers due to MCS's modular process that clarified individual roles and responsibilities, streamlining the development process.

2). Preserved Independency: MCS allowed designers to maintain independency during collaboration, as they could design conversations without considering others' work while still achieving co-operation through MCS components, which was highly agreed by involved designers.

3). Flexibility and Extensibility: Both designers and the developer agreed on and highly appreciated the flexibility and extensibility of chatbots developed using MCS, acknowledging that these benefits to CAs were derived from MCS.

5 CONCLUSIONS AND FUTURE WORK

In this work, we introduce a modular and interactive approach called Micro-Conversation Scheme that fosters collaboration between CA designers and developers while preserving autonomy in creating counseling CAs for psychotherapy and behavioral intervention. This significantly reduces complexity and time consumption during the design and development process. MCS enhances conversational extensibility and flexibility with the help of the Connector and NLG component. The interactive component offers the potential for continuous improvement of CA, involving all stakeholders in the development loop. In the future, we plan to apply MCS to other counseling strategies and carry out evaluation studies to assess its effectiveness. We will quantitatively evaluate the CAs developed using MCS and the effectiveness of collaboration between designers and developers facilitated by MCS.

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